

Exercise Sheet #1: Basic Concepts 1

Lecturer: Prof. Wick

Due: 13 Oct 2025

Exercise 1: Coulomb Force

Two point particles are separated by a distance of 0.60 m and carry a total charge of $200\text{ }\mu\text{C}$. Determine the charge of each of the two particles if they

- a) repel each other with a force of 80 N,
- b) attract each other with a force of 80 N.

Solution

Let the charges be q_1 and q_2 with

$$q_1 + q_2 = Q = 200 \times 10^{-6} \text{ C}.$$

The Coulomb force is

$$F = k \frac{|q_1 q_2|}{r^2},$$

$$k = 8.99 \times 10^9 \text{ N m}^2/\text{C}^2,$$

$$r = 0.60 \text{ m},$$

$$F = 80 \text{ N}.$$

Thus,

$$|q_1 q_2| = \frac{F r^2}{k} \approx 3.20 \times 10^{-9} \text{ C}^2,$$

with

$$q_1 q_2 = q_1 (q - q_1).$$

Case (a): Repulsion

$$q_1 q_2 = +3.20 \times 10^{-9} \Rightarrow x^2 - (2.00 \times 10^{-4})x + 3.20 \times 10^{-9} = 0$$

With

$$ax^2 + bx + c = 0 \Rightarrow x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a},$$

$$q_1 \approx 1.8 \times 10^{-4} \text{ C}, \quad q_2 \approx 1.8 \times 10^{-5} \text{ C}.$$

Case (b): Attraction

$$q_1 q_2 = -3.20 \times 10^{-9} \Rightarrow x^2 - (2.00 \times 10^{-4})x - 3.20 \times 10^{-9} = 0$$

$$q_1 \approx -1.5 \times 10^{-5} \text{ C}, \quad q_2 \approx 2.1 \times 10^{-4} \text{ C}.$$

Reversed signs are also possible.

Exercise 2: Drift Velocity

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Solution

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Exercise 3: Ohm's Law

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Solution

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Exercise 4: Conductance

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Solution

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